Arun Pandey Laudari

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EDUCATION

Lumbini ICT Campus , Tribhuvan University — Expected Graduation: 2026
Bachlore of Computer Application — *Nepal*

Aroma College — Graduated: 2021
Higher Secondary Certificate (Science), GPA: 3.43 — *Nepal*

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, JavaScript, Java, PHP, C, HTML/CSS
- **Frameworks, Libraries & APIs:** Bootstrap, Tailwind CSS, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, NLTK, Hugging Face, GoogleADK, LangChain, LangGraph, FastAPI, Flask, Django, Laravel, Streamlit
- **AI & LLM Technologies:** Agentic AI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, Embeddings, Vector Search
- **Tools & Platforms:** Git, GitHub, Webhooks, ngrok, ClickUp, Draw.io, Docker, Postman, Jupyter Notebook, VS Code, Antigravity
- **Databases & Storage:** PostgreSQL, MySQL, SQLite, Redis, Elasticsearch, Qdrant, ChromaDB, FAISS
- **Concepts:** Data Analysis, Data Visualization, Machine Learning, Deep Learning, Neural Networks, Transformers, NLP, Scalable Backend Development, Background Task Processing (Celery), Search Analytics

EXPERIENCE

Elevate-X Innovation Pvt. Ltd. — Jun 2026 – Sep 2026
AI Engineer & Software Developer — *Bharatpur, Chitwan, Nepal*

- Engineered AI/ML and Generative AI solutions involving Agentic AI, Retrieval-Augmented Generation (RAG), LLM-based applications, and intelligent knowledge systems.
- Developed an enterprise LLM Wiki and knowledge-grounding platform for structured knowledge management, semantic retrieval, contextual understanding, and automated documentation generation.
- Built semantic recommendation and matching pipelines using open-source LLMs, vector databases, embeddings, and intelligent scoring algorithms for candidate and job-profile analysis.
- Designed and developed scalable RESTful backend APIs using Django, Django REST Framework (DRF), Django Ninja, and FastAPI, with PostgreSQL for relational data management and schema design.
- Implemented AI-powered backend workflows integrating LLM inference, RAG pipelines, vector search, prompt engineering, and agent-based architectures for production-oriented applications.
- Developed MCP-based integrations and AI agent workflows to connect language models with external tools, APIs, knowledge sources, and application services.
- Worked with Docker, Git, Redis, Celery, and modern Python-based backend technologies to build maintainable, scalable, and asynchronous software systems.
- Contributed to end-to-end software development, including system architecture, backend engineering, API development, database integration, AI integration, testing, and deployment.

Inspiring Lab Pvt. Ltd. — Nov 2025 – Feb 2026
AI/ML Intern — *Kathmandu, Nepal*

- Developed scalable AI/ML systems with a focus on Agentic AI and Retrieval-Augmented Generation (RAG) for intelligent applications.
- Worked extensively with agent orchestration frameworks such as LangGraph and Google ADK, building multi-agent workflows for adaptive reasoning and task execution.
- Engineered backend services using FastAPI and managed structured data with PostgreSQL for high-performance applications.
- Implemented vector search and retrieval pipelines using Elasticsearch, along with Kibana for monitoring and analytics.
- Designed and deployed an end-to-end AI-driven social media content generation pipeline leveraging for real-time search, trend analysis, and content optimization.
- Built systems capable of generating platform-specific viral content by analyzing engagement patterns and writing styles across social media platforms.
- Integrated Shopify and WooCommerce APIs with webhooks to automate e-commerce workflows and business processes.
- Utilized Redis for conversational memory and Celery for asynchronous background task processing.
- Developed an agent-based conversational system for sales support, enabling intelligent, context-aware customer interactions.

PROJECTS

Rural Health AI – Smart Healthcare Solutions | *Django, LangGraph, ChromaDB, Gemini, SerpAPI, Celery, Redis, PostgreSQL*

- Collaborated in building a healthcare platform using Django's MVT architecture and Django REST Framework, contributing to an Agentic RAG system with a 3-tier intent classification (Heuristic, LLM, Hybrid) for efficient clinical routing.
- Worked on backend development by designing Django models, views, integrating LangGraph agents with ChromaDB for semantic retrieval and SerpAPI fallback to improve response accuracy.
- Contributed to an AI-powered appointment scheduling system, implementing business logic in Django views and optimizing priority handling using a Min-Heap for $O(\log n)$ scheduling efficiency.
- Partnered in developing secure role-based access using Django authentication and middleware, and implemented asynchronous task processing with Celery and Redis for background jobs like notifications and cancellations.

Multi-Tenant Restaurant Management System | *Django, DRF, PostgreSQL, django-tenants, Django Channels, WebSockets, JWT, Docker*

- Engineered a multi-tenant restaurant management backend using Django REST Framework and PostgreSQL schema-based isolation, ensuring secure data separation and independent operations across restaurant tenants.
- Designed and implemented RESTful APIs for table and menu management, order processing, billing, inventory control, and sales analytics, with granular role-based access control across restaurant staff roles.
- Implemented real-time communication using Django Channels and WebSockets to synchronize table status and kitchen order updates, supporting an event-driven Kitchen Display System (KDS).
- Developed secure QR-based ordering with HMAC signature validation, JWT authentication, fiscal-year-based invoice numbering, thermal receipt generation, and concurrency-safe inventory updates.

Agentic RAG ChatBot Using LangGraph | *Python, LangChain, HuggingFace, LangGraph, Gemini, RAG, SerpAPI, FastAPI, React*

- Engineered an end-to-end Agentic RAG pipeline integrating LangGraph agents for adaptive document retrieval and fallback web search via SerpAPI.
- Implemented semantic retrieval using Gemini vector store to enhance contextual awareness and response precision.
- Developed a full-stack system with FastAPI backend and React frontend for real-time query interaction and intelligent answer delivery.

SQL Sage Intelligent DB Agent | *Python, LangChain, HuggingFace, LangGraph, GROQ, RAG, SQLite, Streamlit*

- Developed an intelligent database assistant that converts natural language queries into optimized SQL statements with automatic schema validation.
- Implemented dynamic schema exploration and intelligent query correction to ensure accuracy and reduce SQL errors.
- Built a multi-agent fault-tolerant architecture using LangChain, LangGraph, and RAG for reliable conversational SELECT querying.
- Created both Streamlit and CLI interfaces for real-time database querying and analytics visualization.

Multi-Document Chatbot | *Python, LangChain, LlamaParse, HuggingFace, Chroma, Groq LLM, RAG*

- Built a multi-format document chatbot supporting PDF, DOCX, PPTX, XLSX, Markdown, and TXT files using robust fallback loaders.
- Used HuggingFace embeddings with persistent Chroma vector store for fast semantic retrieval and contextual responses.
- Integrated SelfQueryRetriever with metadata filtering and Groq LLM for accurate and grounded query responses.
- Enabled multi-turn conversational memory with custom prompting to reduce hallucination and improve consistency.

Customer Churn Prediction System Using ANN | *Python, TensorFlow, Keras, Scikit-learn, Streamlit*

- Developed an end-to-end customer churn prediction pipeline using Artificial Neural Networks (ANN) with TensorFlow and Keras.
- Implemented data preprocessing with feature scaling, one-hot encoding, and model persistence for reproducibility.
- Built an interactive Streamlit dashboard for real-time churn prediction and customer segmentation visualization.
- Optimized model performance through hyperparameter tuning, early stopping, and TensorBoard-based monitoring.

Movie Recommendation System | *Python, Flask, Scikit-learn, Pandas, NumPy, TMDB API, Docker*

- Developed a content-based movie recommender using TF-IDF vectorization and cosine similarity.
- Processed TMDB metadata to extract key features (genres, keywords, overview) for accurate recommendation matching.
- Integrated TMDB API for dynamic poster retrieval and implemented autocomplete for enhanced UX.
- Containerized and deployed using Docker and Gunicorn for scalable production serving.

Customer Segmentation Using K-Means Clustering | *Python, Scikit-learn, Pandas, Matplotlib, Seaborn, Streamlit*

- Built an end-to-end customer segmentation system using K-Means clustering to identify actionable behavioral groups.
- Performed data cleaning, feature scaling, and selection for robust clustering outcomes.
- Used Elbow and Silhouette methods for optimal cluster validation and model interpretability.
- Created an interactive Streamlit interface to visualize clusters and generate data-driven marketing insights.

Predicting Diabetes with Logistic Regression | *Python, Scikit-learn, Pandas, Seaborn, Flask*

- Developed a logistic regression-based predictive model for assessing diabetes risk from patient health data.
- Performed data preprocessing, outlier handling, and feature engineering to improve accuracy and stability.
- Trained and validated models with feature scaling and class balancing to enhance prediction reliability.
- Integrated model inference into a Flask-based web app for real-time health risk assessment and interpretability.

ACHIEVEMENTS & CERTIFICATIONS

Complete Data Science, Machine Learning, Deep Learning, and NLP Bootcamp – Udemy course by Krish Naik; covered end-to-end data science pipelines, model deployment, and NLP techniques. *2025*

Machine Learning, Deep Learning, and RAG Frameworks Course – Conducted by Abinash Pant (AI/ML Engineer); focused on building intelligent agents with LangGraph, Agno, and Retrieval-Augmented Generation (RAG) architectures. *2025*

Mentor – Chitwan Utsav 2.0 Hackathon – Guided participants in designing AI-driven solutions and integrating ML models into functional prototypes. *2025*

Hackathon Winner – United Technical College – Recognized for excellence in innovation, technical problem-solving, and teamwork among top participants. *2024*